Domain Notes for Data Analysts/Scientists

(Retail & E-commerce)

01. Overview — What is this domain?

Retail & E-commerce covers the analytics work behind **buying, selling, and delivering products to consumers**, whether through physical stores, online platforms, or both. This domain includes:

- **Brick-and-Mortar Retail** – physical store operations, in-store merchandising, foot traffic analysis
- **E-commerce/Online Retail** – digital storefronts, marketplaces (Amazon, Flipkart), direct-to-consumer (D2C) brands
- **Omnichannel Retail** – integrating online and offline experiences (buy-online-pickup-in-store, ship-from-store)
- **Marketplaces & Platforms** – multi-seller platforms managing catalog, logistics, and seller performance
- **Quick Commerce (Q-commerce)** – ultra-fast delivery models (10-30 minute delivery)
- **Fashion & Apparel, Grocery, Electronics, and other retail verticals** – each with distinct inventory and demand characteristics

Why data analytics/science matters here

1. **Thin margins, high volume** — retail runs on efficiency; small improvements in pricing, inventory, or conversion compound into significant profit impact.
2. **Customer behavior is the product** — understanding what, when, and why customers buy drives merchandising, marketing, and personalization.
3. **Inventory is a double-edged sword** — too much ties up capital and causes markdowns; too little causes stockouts and lost sales. Analytics balances this constantly.
4. **Highly seasonal and promotional** — sales patterns are shaped by festivals, holidays, and discount events, requiring robust demand forecasting.
5. **Massive transactional data** — every click, cart addition, and purchase generates data that can be mined for insight.

Industry context

Retail & e-commerce is less centrally regulated than domains like banking or healthcare, but analysts should be aware of:

- **Consumer Protection Laws** – truthful advertising, pricing transparency (varies by country; e.g., FTC in the US, CCPA for data privacy)
- **Data Privacy Regulations** – GDPR (EU), CCPA (California), India's DPDP Act — govern how customer behavioral data can be collected and used
- **E-commerce-specific regulations** – rules around marketplace seller relationships, return policies, and payment security (PCI-DSS for card data)

Typical business functions a Data Analyst/Scientist supports

- Merchandising & Category Management
- Demand Forecasting & Inventory Planning
- Pricing & Promotions
- Marketing & Customer Analytics
- Supply Chain & Logistics (overlaps with the Supply Chain domain)
- Customer Experience & Personalization
- Store Operations Analytics (for brick-and-mortar/omnichannel)
- Seller/Marketplace Analytics (for platform businesses)

Why this domain is attractive for Data Analysts/Scientists

- Extremely fast-paced with visible, quick feedback loops (a pricing change or campaign shows results within days)
- Rich, high-volume datasets ideal for building strong analytical/ML skills
- Directly ties analytics work to revenue and customer experience — highly visible impact
- Career path: Analyst → Senior Analyst/Category Analytics Lead → Data Scientist → Analytics Manager → Director of Data/Head of Growth

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02. Role of a Data Analyst/Scientist — Day-to-Day & Full Project Lifecycle

A. Day-to-Day Responsibilities (Snapshot)

- Building dashboards to track daily sales, conversion rate, and inventory health

- Running A/B tests on website features, pricing, or promotions
- Building demand forecasting models to guide inventory replenishment
- Analyzing customer segments for targeted marketing campaigns
- Investigating sudden drops/spikes in sales or conversion (root-cause analysis)
- Supporting category managers with pricing and assortment decisions
- Building recommendation engines for personalized product suggestions

B. End-to-End Project Lifecycle (Example: Demand Forecasting for Inventory Planning)

Step 1: Business Problem Understanding

- Meet with Supply Chain/Merchandising leadership to understand the pain point.
  - Example: *"We're experiencing frequent stockouts on fast-moving SKUs during festive season, while overstocking slow-moving items, leading to both lost sales and markdown losses."*
- Translate into an analytical framing:
  - Business ask → *Improve inventory allocation accuracy*
  - Analytical framing → *Build a SKU-level demand forecasting model to predict weekly sales for the next 8-12 weeks, feeding into automated replenishment*
- Define success metrics: forecast accuracy (MAPE/WMAPE), stockout rate reduction, inventory turnover improvement

Step 2: Data Collection & Understanding

- Identify data sources: historical sales transactions, inventory levels, promotional calendar, pricing history, store/warehouse data
- Understand the product hierarchy (category → sub-category → SKU) and how granular the forecast needs to be
- Check data availability across time — need enough historical data to capture seasonality (ideally 2-3 years for annual patterns)
- Identify external factors: holidays, weather (for weather-sensitive categories), competitor promotions if available

Step 3: Data Extraction

- Pull data from POS (Point of Sale) systems, e-commerce order management systems, and warehouse management systems
- Extract at the right granularity — typically SKU x Store x Day (for offline) or SKU x Day (for online)
- Join with promotional calendar data (which weeks had discounts, and how deep)
- Handle new product launches carefully (no historical data — requires different treatment, like using similar/analogous product data)

Step 4: Data Cleaning & Preprocessing

- Handle missing sales data due to stockouts (a stockout day shows zero sales, but that's NOT the same as zero demand — this is a critical distinction requiring demand "unconstraining" techniques)
- Remove or flag outliers (e.g., bulk B2B orders that distort typical retail demand patterns)
- Standardize SKU identifiers across systems (online and offline systems may use different product codes)
- Handle returns/cancellations appropriately (net sales vs gross sales)

Step 5: Exploratory Data Analysis (EDA)

- Analyze sales trends by category, seasonality patterns (weekly, monthly, festival-driven spikes)
- Identify the impact of past promotions on sales lift (and any "cannibalization" — a promotion pulling forward future sales rather than creating net new demand)
- Segment products by demand pattern (fast-moving vs slow-moving, seasonal vs steady, new vs mature)
- Visualize demand variability (highly volatile SKUs need different forecasting treatment than stable ones)

Step 6: Feature Engineering

- Create time-based features: day of week, week of year, month, holiday flags, days to/from major festivals
- Create promotional features: discount depth, promotion type, days since last promotion
- Create lag features: sales in the same week last year, rolling averages of recent weeks

- Create product/store features: price point, category, store format/size, store location demographics
- Incorporate external signals where relevant: weather data, local events, competitor pricing (if available)

Step 7: Model Building

- Choose forecasting approach based on data characteristics and forecast granularity:
  - **Statistical methods**: ARIMA/SARIMA, Exponential Smoothing (Holt-Winters) for stable, well-behaved series
  - **Machine Learning methods**: Gradient Boosting (XGBoost, LightGBM) treating forecasting as a regression problem with engineered time/lag features — popular for handling many SKUs with irregular patterns
  - **Deep Learning methods**: LSTM, Temporal Fusion Transformer, or Amazon's DeepAR for large-scale, cross-series learning (useful when many related SKUs can share learned patterns)
  - **Hierarchical forecasting** – ensuring forecasts are consistent when aggregated from SKU level up to category/company level
- For new products without history, use analog forecasting (based on similar existing products) or judgmental overrides from merchandising teams

Step 8: Model Validation

- Use time-based train/test splits (never randomly split time-series data) — train on historical periods, validate on more recent held-out periods
- Evaluate using forecast accuracy metrics: **MAPE (Mean Absolute Percentage Error)**, **WMAPE (Weighted MAPE)**, **Bias** (systematic over/under-forecasting)
- Backtest across multiple historical periods, including past festive seasons, to ensure the model handles peak demand periods well
- Validate with merchandising/supply chain teams — do the forecasts align with business intuition and known upcoming events?

Step 9: Bias & Edge Case Review

- Check forecast performance specifically for new product launches, promotional periods, and high-volatility SKUs (these are typically the hardest and most business-critical to get right)

- Ensure the model doesn't systematically under-forecast fast-growing products or over-forecast declining ones
- Review extreme/edge cases with category managers for sanity checks

Step 10: Deployment

- Integrate the forecasting model into the inventory planning/replenishment system, so forecasts automatically feed into purchase order and stock allocation decisions
- Build a dashboard for planners to review, override (with documented reasons), and approve forecasts before they trigger automated actions
- Set up automated retraining/refresh schedules (e.g., weekly re-forecast as new sales data comes in)

Step 11: Monitoring & Maintenance

- Track forecast accuracy (MAPE/bias) on an ongoing basis, by category and by forecast horizon (near-term forecasts should be more accurate than far-term)
- Monitor for forecast drift during unusual events (supply chain disruptions, sudden demand shifts, new competitor entry)
- Set up alerts for large forecast errors requiring planner review
- Periodically retrain models with fresh data and re-evaluate feature relevance

Step 12: Reporting & Stakeholder Communication

- Present forecast accuracy improvements and business impact (reduced stockouts, improved inventory turnover, markdown reduction) to supply chain and merchandising leadership
- Provide category-level insights back to merchandising teams (e.g., "Category X shows strong weekend demand patterns not currently reflected in store staffing")

C. This Lifecycle Applies (with Variations) to Other Retail Use Cases

- **Pricing/Markdown Optimization** – similar data pipeline but target is optimal price point rather than demand quantity, often combined with price elasticity modeling
- **Customer Churn/Retention Models** – shifts focus from product-level to customer-level data (purchase history, engagement)
- **Recommendation Systems** – uses similar behavioral data but focuses on personalization rather than aggregate demand

- **Store Network Planning** – works at a much higher level (store/region) with longer planning horizons (months to years)

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03. Key Metrics & KPIs

Sales & Revenue Metrics

Metric	Meaning
Gross Merchandise Value (GMV)	Total value of goods sold through a platform (before returns/discounts)
Net Revenue	Revenue after returns, discounts, and cancellations
Average Order Value (AOV)	Average amount spent per transaction
Same-Store Sales Growth (SSSG)	Revenue growth from stores open for over a year (excludes new store openings)
Sales per Square Foot	Store productivity measure (physical retail)

Customer & Conversion Metrics

Metric	Meaning
Conversion Rate	% of visitors/shoppers who complete a purchase
Cart Abandonment Rate	% of shoppers who add items to cart but don't complete checkout
Customer Acquisition Cost (CAC)	Cost to acquire a new customer
Customer Lifetime Value (CLV/LTV)	Total projected revenue/profit from a customer over their relationship
Repeat Purchase Rate	% of customers who make more than one purchase
Churn Rate	% of customers who stop purchasing over a given period
Net Promoter Score (NPS)	Customer satisfaction/loyalty measure

Inventory & Supply Chain Metrics

Metric	Meaning
Inventory Turnover Ratio	How many times inventory is sold and replaced over a period
Stockout Rate	% of time a SKU is unavailable when demand exists
Sell-Through Rate	% of received inventory that has been sold in a given period
Days of Inventory Outstanding (DIO)	Average number of days inventory sits before being sold
Markdown/Discount Rate	% of sales sold at a discounted price
Forecast Accuracy (MAPE/WMAPE)	How close demand forecasts are to actual sales

Marketing & Digital Metrics

Metric	Meaning
Click-Through Rate (CTR)	% of ad/email viewers who click through
Return on Ad Spend (ROAS)	Revenue generated per unit of advertising spend
Cost per Acquisition (CPA)	Marketing cost to acquire one converting customer
Email Open Rate/Engagement Rate	Effectiveness of email marketing campaigns

Pricing & Profitability Metrics

Metric	Meaning
Gross Margin	Revenue minus cost of goods sold, as a % of revenue
Price Elasticity of Demand	Sensitivity of demand to price changes
Contribution Margin	Profit contribution after variable costs, used in pricing/promotion decisions

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04. Common Datasets & Data Sources

Internal/Transactional Data Sources

Source System	Typical Data
POS (Point of Sale) Systems	In-store transaction data, timestamps, payment methods
E-commerce Platform/Order Management System (OMS)	Online orders, cart data, checkout funnel data
Inventory/Warehouse Management System (WMS)	Stock levels, replenishment data, warehouse locations
CRM Systems	Customer profiles, loyalty program data, service interactions
Web/App Analytics (Google Analytics, Adobe Analytics)	Clickstream data, page views, session data, funnel drop-offs
Marketing Platforms	Campaign performance, email/SMS engagement data

External Data Sources

Source	Typical Data
Weather Data Providers	Historical/forecast weather (relevant for weather-sensitive categories like apparel, beverages)
Competitor Pricing Data (via web scraping or data providers)	Competitor product prices for benchmarking
Social Media/Review Platforms	Customer sentiment, product reviews, influencer engagement
Economic/Demographic Data	Regional income levels, population data for store location planning

Typical Fields/Attributes an Analyst Works With

Transaction Data:

- Order ID, customer ID, SKU, quantity, unit price, discount applied, payment method, timestamp, channel (online/offline)

Product Data:

- SKU, category, sub-category, brand, cost price, selling price, size/color variants

Customer Data:

- Customer ID, demographics (age, location, gender if available), purchase history, loyalty tier, acquisition channel

Clickstream/Web Data:

- Session ID, page views, time on page, cart additions, checkout steps completed, device type, traffic source

Inventory Data:

- SKU, store/warehouse location, stock on hand, in-transit inventory, reorder point

Data Characteristics Specific to Retail & E-commerce

- **High seasonality** — sales patterns shift dramatically around holidays, festivals, and sales events
- **Sparse data for long-tail products** — many SKUs sell infrequently, making individual-level forecasting hard
- **Cold-start problem** — new products/customers have no historical data, requiring special handling
- **High cardinality** — millions of SKU-store or SKU-customer combinations in large retailers
- **Fast-changing catalog** — products are frequently added, discontinued, or re-priced

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05. Tools & Technologies

Programming Languages

- **SQL** – essential for querying transactional databases and data warehouses
- **Python** – Pandas, NumPy, Scikit-learn, XGBoost/LightGBM, Prophet for forecasting and analysis
- **R** – used in some analytics teams for statistical modeling and visualization

Data Warehouses & Big Data

- **Snowflake, Google BigQuery, Amazon Redshift** – common data warehouse platforms for retail-scale data
- **Apache Spark/PySpark** – for processing large transactional/clickstream datasets
- **Hadoop** – used in some legacy large-scale retail data infrastructure

Visualization & BI Tools

- **Tableau, Power BI, Looker** – for sales, inventory, and marketing dashboards
- **Google Analytics/Adobe Analytics** – for web/app behavior analysis
- **Excel** – still widely used for ad-hoc analysis and category management reporting

Forecasting & ML Tools

- **Prophet (Facebook/Meta)** – popular for business-friendly time-series forecasting
- **Scikit-learn, XGBoost, LightGBM** – for ML-based demand forecasting and classification tasks
- **Deep learning frameworks (TensorFlow/PyTorch)** – for advanced forecasting (DeepAR, Temporal Fusion Transformer) at scale

Experimentation & Personalization

- **A/B testing platforms** (Optimizely, VWO, or in-house frameworks) – for testing website/app changes
- **Recommendation engine frameworks** – collaborative filtering libraries, or cloud services like AWS Personalize

Cloud & MLOps

- **AWS, Azure, GCP** – for scalable data processing, storage, and ML model deployment
- **Airflow/dbt** – for data pipeline orchestration and transformation
- **MLflow, Kubeflow** – for model versioning and deployment at scale

E-commerce/Retail-Specific Platforms

- **Shopify, Magento, Salesforce Commerce Cloud** – common e-commerce platform backends (source of transactional data)
- **SAP, Oracle Retail** – enterprise retail management systems used by large traditional retailers

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06. Real-World Use Cases

Merchandising & Demand Planning

- **Demand Forecasting** – as detailed in the lifecycle example above
- **Assortment Planning** – deciding which products to stock in which stores/regions based on local demand patterns
- **New Product Launch Forecasting** – predicting demand for products with no sales history

Pricing & Promotions

- **Dynamic Pricing** – adjusting prices in real-time based on demand, competition, and inventory levels
- **Markdown Optimization** – determining optimal discount timing/depth to clear excess inventory while maximizing revenue
- **Promotion Effectiveness Analysis** – measuring true incremental lift from promotions (accounting for cannibalization and pull-forward effects)

Customer Analytics & Personalization

- **Customer Segmentation (RFM Analysis)** – grouping customers by Recency, Frequency, Monetary value for targeted engagement
- **Recommendation Engines** – "customers who bought this also bought" and personalized homepage/email recommendations
- **Churn Prediction** – identifying customers likely to stop purchasing, enabling proactive retention campaigns
- **Customer Lifetime Value (CLV) Modeling** – predicting long-term value to prioritize marketing spend

Marketing Analytics

- **Marketing Mix Modeling (MMM)** – measuring the impact of different marketing channels on sales
- **Attribution Modeling** – determining which marketing touchpoints contributed to a conversion

- **Campaign Response Modeling** – predicting which customers are likely to respond to a specific campaign

Supply Chain & Operations (Overlaps with Supply Chain Domain)

- **Inventory Optimization** – balancing stock levels against demand uncertainty and holding costs
- **Store Clustering** – grouping similar stores for standardized planning and benchmarking
- **Last-Mile Delivery Optimization** – especially relevant for e-commerce and quick-commerce players

Fraud & Risk (E-commerce Specific)

- **Payment Fraud Detection** – identifying fraudulent transactions in real-time
- **Fake Review/Rating Detection** – identifying manipulated product reviews on marketplaces
- **Return Fraud Detection** – identifying abuse of return policies (e.g., "wardrobing")

Example Interview-Style Problem Statements

- *"Build a demand forecasting model for a retailer's top 100 SKUs ahead of the holiday season."*
- *"How would you measure the true incremental impact of a 20%-off promotion, accounting for cannibalization?"*
- *"Design a recommendation system for an e-commerce homepage that balances relevance and product discovery."*
- *"Our cart abandonment rate has increased 5% month-over-month — how would you investigate the cause?"*

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07. ML & Statistical Techniques

Forecasting Techniques

- **ARIMA/SARIMA** – classical time-series forecasting for stable, well-behaved demand patterns
- **Exponential Smoothing (Holt-Winters)** – captures trend and seasonality, simple and interpretable

- **Prophet** – business-friendly forecasting tool handling holidays/seasonality well with minimal tuning
- **Gradient Boosting (XGBoost, LightGBM) for Forecasting** – treating time-series forecasting as a supervised regression problem using lag/rolling features
- **Deep Learning (LSTM, DeepAR, Temporal Fusion Transformer)** – for large-scale forecasting across many related SKUs simultaneously

Customer Analytics Techniques

- **RFM Analysis** – rule-based segmentation using Recency, Frequency, Monetary value
- **Clustering (K-Means, Hierarchical Clustering)** – for behavioral/demographic customer segmentation
- **Survival Analysis** – for modeling time-to-churn
- **Uplift Modeling** – measuring the incremental effect of a marketing intervention on individual customers, rather than just correlation

Recommendation System Techniques

- **Collaborative Filtering (User-based, Item-based)** – classic recommendation approach based on user-item interaction patterns
- **Matrix Factorization (SVD, ALS)** – decomposing user-item interactions into latent factors
- **Content-Based Filtering** – recommending based on product attributes/similarity
- **Hybrid & Deep Learning Recommenders** – combining collaborative and content signals, increasingly using neural embeddings

Pricing & Promotion Techniques

- **Price Elasticity Modeling (Regression-based)** – estimating how demand responds to price changes
- **Causal Inference Methods (Difference-in-Differences, Synthetic Control)** – measuring true incremental impact of pricing/promotional changes
- **A/B Testing & Multi-Armed Bandits** – for testing and optimizing pricing/promotional strategies in real-time

Classification & Propensity Models

- **Logistic Regression, Random Forest, XGBoost** – for churn prediction, campaign response modeling, fraud detection
- **Association Rule Mining (Apriori, FP-Growth)** – for "frequently bought together" analysis (market basket analysis)

Marketing Mix & Attribution Modeling

- **Bayesian Marketing Mix Models (MMM)** – estimating channel-level contribution to sales while accounting for diminishing returns and carryover effects (adstock)
- **Multi-Touch Attribution Models** – distributing conversion credit across multiple marketing touchpoints (linear, time-decay, Markov chain-based attribution)

Statistical Testing

- **A/B Testing (T-tests, Chi-Square Tests)** – for website/app feature testing
- **Sequential Testing/Bayesian A/B Testing** – for faster experiment decision-making in high-traffic e-commerce environments

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08. Resources

Public Datasets for Practice

- **Kaggle: "Instacart Market Basket Analysis"** – customer purchase pattern and basket analysis dataset
- **Kaggle: "Online Retail Dataset (UCI)"** – classic transactional dataset for RFM/customer segmentation practice
- **Kaggle: "Walmart Recruiting - Store Sales Forecasting"** – demand forecasting practice dataset
- **Kaggle: "Brazilian E-Commerce Public Dataset (Olist)"** – comprehensive e-commerce dataset covering orders, reviews, and logistics
- **M5 Forecasting Competition Dataset (Kaggle)** – large-scale hierarchical retail demand forecasting dataset from Walmart

Recommended Courses

- Retail Analytics courses (Coursera – various universities)
- Marketing Analytics Specialization (Coursera – University of Virginia)

- Demand Forecasting and Inventory Management courses (Coursera/edX)
- Customer Analytics courses (Coursera – University of Pennsylvania/Wharton)

Books

- *"Big Data and the Retail Ecosystem"* – various retail analytics practitioner references
- *"Forecasting: Principles and Practice"* – Hyndman & Athanasopoulos (free online, the standard time-series forecasting reference)
- *"Retail Analytics: The Secret Weapon"* – Emmett Cox
- *"Competing on Analytics"* – Davenport & Harris (broader business analytics reference with strong retail examples)

Communities & Blogs

- Kaggle competition forums (search "retail", "demand forecasting", "recommendation systems")
- Towards Data Science (Medium) – search retail/e-commerce analytics articles
- RetailDive, RetailWire – industry news and trend analysis
- LinkedIn groups: "Retail Analytics Professionals", "E-commerce Data Science"

GitHub Repositories for Reference

- Search GitHub for: demand-forecasting, recommendation-system, market-basket-analysis, customer-segmentation-rfm for open-source implementation examples

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This completes the Retail & E-commerce domain notes.